

The Double Angle Identities (Homework)

Prove that the following equations are true by converting the left side to the right.

$$\textcircled{1} \frac{2\cos^2 x}{\sin(2x)} = \cot x \quad \textcircled{2} \sin^2 A - \cos(2A) = 2\sin^2 A - \cos^2 A$$

$$\textcircled{3} \frac{\cos(2B) - 1}{2} = -\sin^2 B$$

$$\textcircled{4} \frac{\sin(2\theta)}{2\cos\theta} = \sin\theta$$

$$\textcircled{5} \frac{\sin(2W)}{4\cos^2 W} = \frac{\tan W}{2} \quad \textcircled{6} \frac{\cos(2\beta)}{\sin^2 \beta} = 2\cot^2 \beta - \csc^2 \beta$$

$$\textcircled{7} \frac{\sin A + 1}{\sin(2A)} = \frac{\sec A + \csc A \sec A}{2} \quad \textcircled{8} \frac{\sin X + \cos X}{\cos(2X)} = \frac{1}{\cos X - \sin X}$$

⑨ Solve if $0 \leq \theta \leq 360^\circ$

$$\cos(2\theta) - \cos^2\theta = -1$$

⑩ Solve if $0 < x < 360^\circ$

$$\sin x = 3 \sin(2x)$$

⑪ Solve if $0 < \theta \leq 360^\circ$

$$\cos(2\theta) + 4 = -2\cos^2\theta + 6$$

⑫ Solve if $0 \leq \alpha < 360^\circ$

$$0 = \sin(2\alpha) - 2\sin^2\alpha$$

⑬ Solve if $0 \leq W \leq 360^\circ$

Hint: There is no solution, but you must show why.

$$\frac{\sin(2W)}{2\cos W} = \sin^2 W - 6$$

⑭ Solve if $0 < X < 360^\circ$

$$8\cos^2 X - 3\cos X - 5 = \cos(2X) + 2\cos X$$

Homework Answers

$$\textcircled{1} \frac{2\cos^2 x}{\sin(2x)} = \cot x \quad \textcircled{2} \sin^2 A - \cos(2A) = 2\sin^2 A - \cos^2 A$$

$$\frac{2\cos^2 x}{2\sin x \cos x} =$$

$$\frac{\cos x}{\sin x} =$$

$$\cot x =$$



$$\textcircled{3} \frac{\cos(2B) - 1}{2} = -\sin^2 B$$

$$\frac{1 - 2\sin^2 B - 1}{2} =$$

$$\frac{-2\sin^2 B}{2} =$$

$$-\sin^2 B =$$



$$\textcircled{4} \frac{\sin(2\theta)}{2\cos\theta} = \sin\theta$$

$$\frac{2\sin\theta\cos\theta}{2\cos\theta} =$$

$$\sin\theta =$$



$$\textcircled{5} \frac{\sin(2w)}{4\cos^2 w} = \frac{\tan w}{2} \quad \textcircled{6} \frac{\cos(2\beta)}{\sin^2 \beta} = 2\cot^2 \beta - \csc^2 \beta$$

$$\frac{2\sin w \cos w}{4\cos^2 w} =$$

$$\frac{\sin w}{2\cos w} =$$

$$\frac{\tan w}{2} =$$



$$\frac{2\cos^2 \beta - 1}{\sin^2 \beta} =$$

$$\frac{2\cos^2 \beta}{\sin^2 \beta} - \frac{1}{\sin^2 \beta} =$$

$$2\cot^2 \beta - \csc^2 \beta =$$



$$\textcircled{7} \frac{\sin A + 1}{\sin(2A)} = \frac{\sec A + \csc A \sec A}{2} \quad \textcircled{8} \frac{\sin X + \cos X}{\cos(2X)} = \frac{1}{\cos X - \sin X}$$

$$\frac{\sin A + 1}{2 \sin A \cos A} =$$

$$\frac{\sin A}{2 \sin A \cos A} + \frac{1}{2 \sin A \cos A} =$$

$$\frac{1}{2 \cos A} + \frac{1}{2 \sin A \cos A} =$$

$$\frac{\sec A}{2} + \frac{\csc A \sec A}{2} =$$

$$\frac{\sec A + \csc A \sec A}{2} =$$



$$\frac{\sin X + \cos X}{\cos^2 X - \sin^2 X} =$$

$$\frac{\sin X + \cos X}{(\cos X + \sin X)(\cos X - \sin X)} =$$

$$\frac{1}{\cos X - \sin X} =$$



$$\textcircled{9} \theta = 90^\circ, 270^\circ$$

$$\textcircled{10} X = 180^\circ \text{ or } X \approx 80.41^\circ, 279.59^\circ$$

$$\textcircled{11} \theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$

$$\textcircled{12} \alpha = 0^\circ, 45^\circ, 180^\circ, 225^\circ$$

$$\textcircled{13} \sin W \neq 3, \sin W \neq -2, \text{ No solution}$$

$$\textcircled{14} X = 120^\circ, 240^\circ$$